Assignment 2: X-ray Powder Diffraction - instrument simulation, crystallite size, and the signals view

Course: Nanotechnology & Quantum Physics | **Topic:** XRD instrument simulation, domain size, the Fourier description of line broadening

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Built with: Prime Agent (the AI assistant used in this session); the whole simulator is in Appendix A.

Item	What it is
Simulator	Cu K-alpha tube, divergence and Soller slits, flat specimen on a theta/2theta goniometer, receiving slit, optional monochromator, detector
Phases	Si, Al, Cu, alpha-Fe, NaCl, MgO, ZnO, TiO2 rutile, TiO2 anatase
Method	Bragg positions from the cell, structure factors with Waasmaier-Kirfel form factors [6], powder multiplicities, Lorentz-polarisation, Debye-Waller, the K-alpha doublet, Caglioti resolution, Scherrer broadening, Poisson counting statistics
Repository	https://github.com/OBODA0/quantum-physics

Abstract

A Bragg-Brentano powder diffractometer was built as a Python model: Cu K-alpha radiation, a flat specimen on a theta/2theta goniometer, structure factors for nine phases, a pseudo-Voigt line shape and Poisson counting statistics. The model reproduces the published patterns of diamond, FCC, BCC, rocksalt, wurtzite and tetragonal structures, and of one structure at crystallite sizes from 5 nm to 500 nm. Measuring that size series shows that the domain size changes the width of every line as 1/D, changes the height in proportion, leaves the integrated area and the Bragg positions unchanged, and broadens high-angle lines more than low-angle ones. All four observations follow from one statement: the finite crystallite is a window on the lattice, so the ideal Bragg impulse train is convolved with the transform of that window, and the instrument is a second convolution in series. The 1/cos(theta) is the Jacobian between reciprocal space and the goniometer angle. The measured widths agree with that prediction without fitting.

1. What this report does

The brief asks for three things. Item 1 is a working X-ray diffractometer simulator and screenshots showing that it reproduces the patterns of materials with different crystal structures and different crystallite sizes. Item 2 asks what happens to the data when the crystal domain size changes. Item 3 asks for the reason, in the language of signals and systems, with credible sources.

The report is organised in the order of the questions. Section 2 describes the instrument and the model that stands behind every number in it; section 3 answers item 1 with the figures; section 4 answers item 2 with a measured size study; section 5 answers item 3 with the Fourier and linear-systems argument and checks it against the measurements. Section 6 records what was verified, what the uncertainty is and what the model deliberately does not do. The code is Appendix A.

One convention is used throughout. The brief speaks of the *size of the crystalline unit* and of the *crystal domain size*; both mean the size of the coherently diffracting domain, written D here, which is the quantity that appears in the Scherrer relation. The word *crystallite size* is used from here on.

2. The instrument and the model

The simulator models a Bragg-Brentano powder diffractometer (Figure 1). A copper tube supplies the K-alpha doublet; a divergence slit and Soller slits define the beam; the flat specimen sits on a theta/2theta goniometer; a receiving slit and an optional monochromator sit in front of the detector. The measured variable is 2theta, and the specimen sits at theta, so a reflection appears where

```
2 d sin(theta) = n lambda    (1)
```

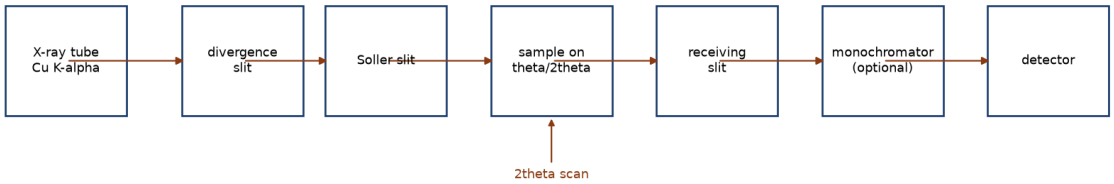


Figure 1: the chain the simulator models. Each box corresponds to a module of the code in Appendix A.

2.1 Positions

The interplanar spacing follows from the cell. For the cubic, tetragonal and hexagonal settings the simulator implements

```
1/d^2 = (h^2 + k^2 + l^2)/a^2    (cubic)

1/d^2 = (h^2 + k^2)/a^2 + l^2/c^2    (tetragonal)

1/d^2 = 4/3 (h^2 + hk + k^2)/a^2 + l^2/c^2    (hexagonal)
```

A reflection whose spacing is shorter than lambda/2 cannot be reached by the radiation and is dropped rather than clamped. The same code is also used to refine a lattice parameter from indexed peaks by a least-squares fit through the origin, which is how the cells in Table 1 were re-checked.

2.2 Intensities

The amplitude scattered by one cell is the structure factor

```
F(hkl) = sum_j occ_j f0_j(s) exp(-B_j s^2) exp(2 pi i (h x_j + k y_j + l z_j)),
```

with s = sin(theta)/lambda = 1/(2d) by Bragg's law, f0 the atomic form factor and B_j the Debye-Waller coefficient of site j [2]. The form factors are the Waasmaier-Kirfel parameterisation [6], interpolated from a table generated once from xraydb and shipped with the code; f0(0) reproduces the electron count, which the test suite checks. The intensity of a powder line is then

```
I(hkl) = |F|^2 m LP(2theta),    LP = (1 + cos^2 2theta)/(sin^2 theta cos theta),
```

where m is the multiplicity of the plane family, computed by counting the orbit of (hkl) under the Laue class rather than read from a table, and LP is the Lorentz-polarisation factor for an unpolarised tube [3]. Where a monochromator is switched on, the polarisation term becomes 1 + cos^2 2theta cos^2 2theta_m. The absolute prefactor of a Bragg-Brentano scan - electron radius, beam area, specimen-to-detector distance, cell volume - is the same for every reflection and cancels in the normalised pattern, which is why the simulation reports relative intensities, as a diffractometer's software does. For the same reason the flat-plate absorption factor A = 1/(2 mu) is independent of angle for a thick specimen and cancels as well; it is computed in the code but not applied.

Systematic absences are not tabulated anywhere in the code. The diamond 222, the FCC 100 and 110 and the BCC 100 and 111 come out as zero because the phases of the basis cancel in the sum above, which is a check that the basis is right rather than a rule applied on top of it.

2.3 The scan

Each line is drawn as a pseudo-Voigt profile, a mixture of a unit-area Gaussian and a unit-area Lorentzian weighted by eta. The instrumental width follows the Caglioti form, and a crystallite of size D adds a Scherrer width:

```
FWHM^2 = U tan^2 theta + V tan theta + W    (instrument)

beta = K lambda/(D cos theta)    (crystallite size, K = 0.9)
```

The two widths are combined in quadrature, each reflection is drawn twice at its own K-alpha1 and K-alpha2 positions with weights 1 and 0.497, and the expected counts are turned into observed counts by a Poisson draw. A dead time compresses strong lines when it is switched on. The scan therefore carries the two features that matter for item 2: a width that depends

on the specimen, and counting noise that does not.

The instrument constants used everywhere in this report are: K-alpha1 1.540598 A, K-alpha2 1.544426 A, Caglioti U = 0.004, V = -0.002, W = 0.003 deg^2, eta = 0.5, and a scan ceiling of 150 deg. They are modelling choices, documented as such in data/SOURCES.md; the cell constants and their records are listed in Appendix B.

3. Item 1: the design, and screenshots of it working (25%)

The instrument of Figure 1 is implemented as a small Python package. Each stage is a module, so a figure can be traced back to the code that produced it:

Module	Responsibility
lattice.py	d-spacings, Bragg angles, lattice refinement
atoms.py, structure.py	form factors, structure factor, multiplicities, LP
sources.py, geometry.py	Cu K-alpha lines, flat-plate geometry, displacement
samples.py	the nine phases: cell, basis, provenance
profile.py, detector.py	pseudo-Voigt, Caglioti, Scherrer, Poisson, dead time
diffractometer.py	the reflection list and the scan itself
analysis.py	peak finding, widths, indexing, Scherrer, Williamson-Hall, phase ID
plot.py, cli.py	rendering and the command line interface

Table 1: the module map. Appendix A carries the source of every module, as item 1 requires.

3.1 Varying the crystal structure

Figure 2 is one run of the simulator on nine phases, with the instrument settings held fixed: 20-90 deg, Cu K-alpha, 20 000 counts at the strongest line, no counting noise. Everything that differs between the panels is crystallography. The four cubic metals differ by their lattice and their basis: aluminium and copper are FCC, so 100 and 110 are absent and 111 is the strongest line; alpha-iron is BCC, so 100 and 111 are absent and 110 leads; silicon carries a two-atom diamond basis on the same FCC lattice, which removes 222 as well and leaves 111 dominant. Rock salt and periclase share the rocksalt structure: their 111 is weak because the two form factors almost cancel at that spacing, while 200 is strong, which is a direct demonstration that intensity comes from the phase sum and not from the cell. Zincite is hexagonal wurtzite, so the cubic degeneracies are lifted and 101 becomes the strongest line. The two titania polymorphs share a formula and differ in structure: rutile is tetragonal with 110 strongest, anatase is body-centred tetragonal with 101 strongest.

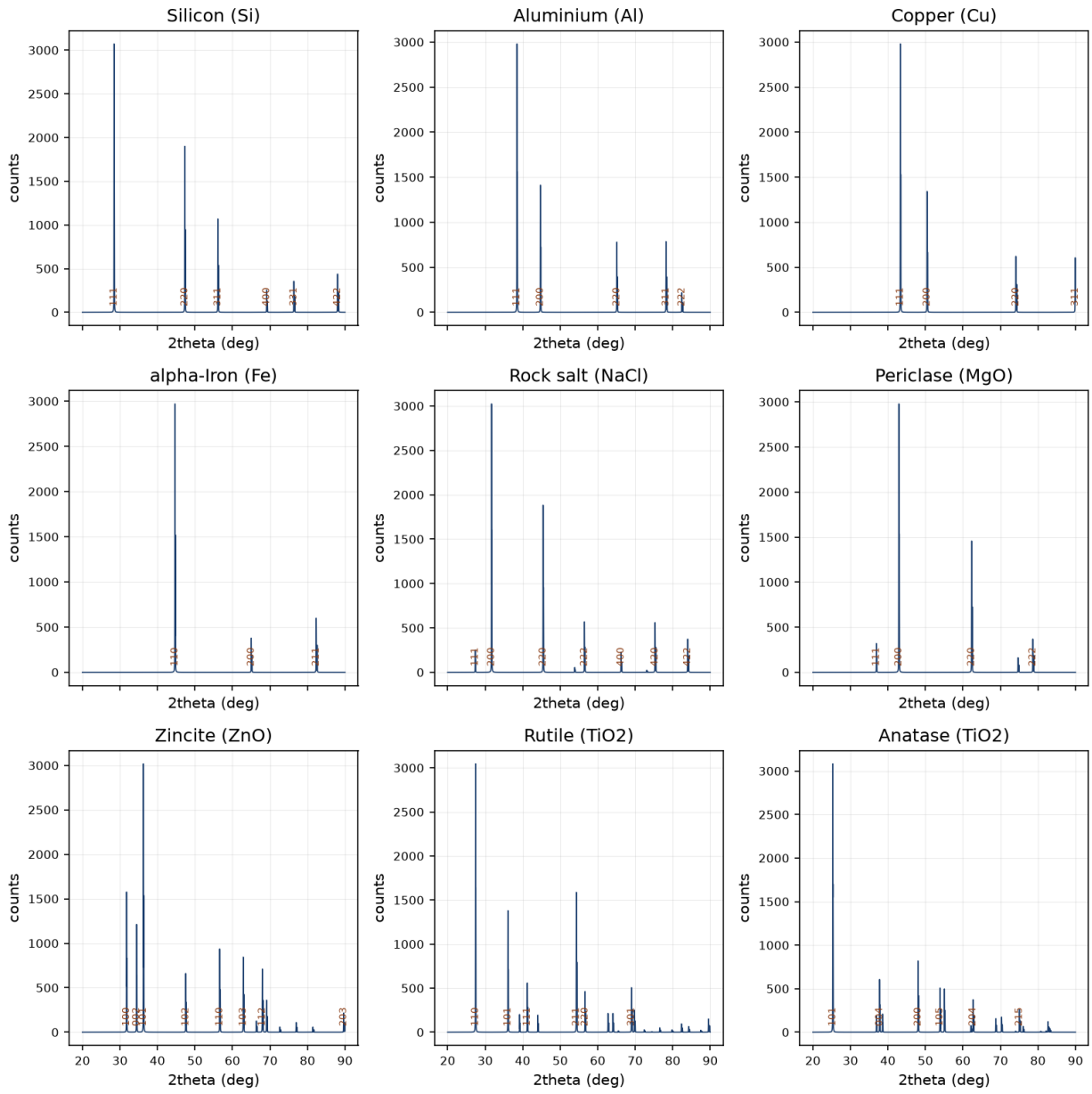


Figure 2: simulated patterns of nine phases under one set of instrument settings. Labels mark the reflections the model predicts.

Key	Phase	Space group	Cell (A)	Strongest line
si	Silicon (Si)	Fd-3m (227)	a = 5.4311	111 at 28.44 deg
al	Aluminium (Al)	Fm-3m (225)	a = 4.0495	111 at 38.47 deg
cu	Copper (Cu)	Fm-3m (225)	a = 3.6149	111 at 43.32 deg
fe	alpha-Iron (Fe)	Im-3m (229)	a = 2.8665	110 at 44.67 deg
nacl	Rock salt (NaCl)	Fm-3m (225)	a = 5.6402	200 at 31.70 deg
mgo	Periclase (MgO)	Fm-3m (225)	a = 4.2112	200 at 42.92 deg
zno	Zincite (ZnO)	P63mc (186)	a = 3.2494, c = 5.2038	101 at 36.26 deg
tio2-rutile	Rutile (TiO2)	P42/mnm (136)	a = 4.5937, c = 2.9587	110 at 27.44 deg
tio2-anatase	Anatase (TiO2)	I41/amd (141)	a = 3.7845, c = 9.5143	101 at 25.31 deg

Table 2: the nine phases, their space groups and cells, and the strongest line the model predicts. Cell constants and their records are in Appendix B and in data/SOURCES.md; every row is checked against its published strongest reflection by the test suite.

3.2 Varying the size of the crystalline unit

Figure 3 repeats one phase, one reflection and one instrument setting while only the crystallite size changes. The position does not move, the peak widens and its height falls. The right-hand panel normalises each curve to its own maximum so that the width can be compared without the height.

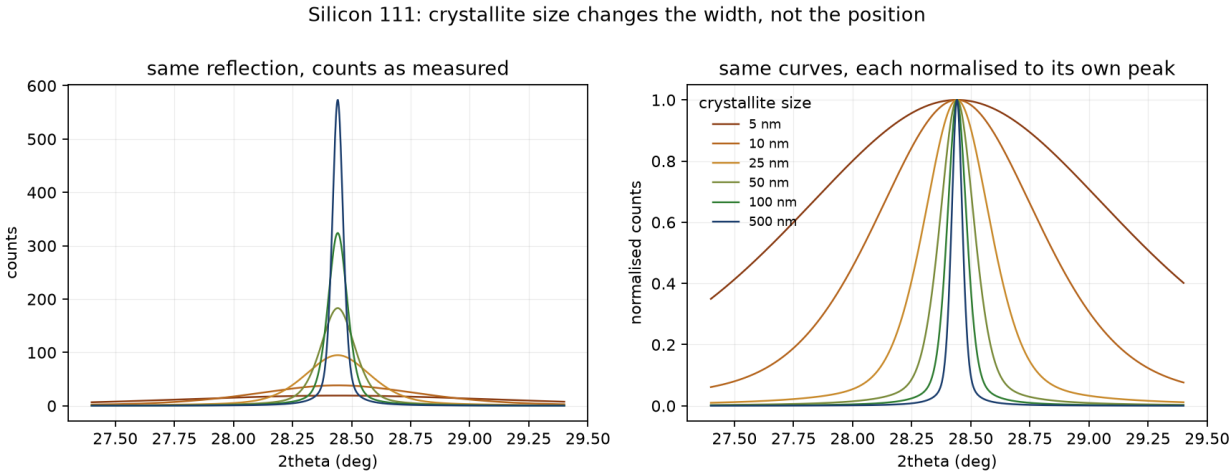


Figure 3: silicon 111 at 5, 10, 25, 50, 100 and 500 nm. Left: counts as measured. Right: each curve normalised to its own peak. The K-alpha2 line is switched off so that the doublet does not confuse the width.

The screenshots are written directly from the simulator to image files by `report/make_figures.py`, which is part of the same repository as the code. They are not retouched: the curves in Figures 2 and 3 are the arrays the scan returns, plotted without smoothing.

3.3 How the model compares with published patterns

A simulation is only useful if its numbers are the numbers a real instrument would give. Table 3 compares the relative intensities the model produces with the values printed on standard powder cards for five of the phases. The reflections are the strongest ones; the model normalises each pattern to its own 111 or 200 line, as the cards do.

Phase	Reflections	Model	Published card
Silicon	111 / 220 / 311 / 400	100 / 64 / 37 / 10	100 / 55 / 30 / 5
Copper	111 / 200 / 220 / 311	100 / 46 / 24 / 27	100 / 46 / 20 / 17
Rock salt	111 / 200 / 220	8 / 100 / 64	13 / 100 / 55
Zincite	100 / 002 / 101	52 / 40 / 100	57 / 44 / 100
Rutile	110 / 101 / 211	100 / 45 / 55	100 / 50 / 60

Table 3: relative intensities, model against published powder cards. The published values are peak heights read from printed cards; the model returns integrated intensities.

The agreement is good on the low-angle lines and drifts high at the back of the scan: silicon 400 at 10 against 5, copper 311 at 27 against 17, rutile 110/101/211 at 100/45/55 against 100/50/60. Two effects explain the direction. First, a printed card lists peak heights while the model returns integrated intensities, and the instrument width grows with angle, so the same area gives a lower height at high angle. Second, the model applies one Debye-Waller coefficient of 0.5 A^2 to every site and no absorption correction, both of which suppress high-angle intensity in a real specimen more than in a single-parameter model. The discrepancy is recorded rather than tuned, because fitting B per site would be fitting, not physics.

4. Item 2: what changes when the domain size changes (25%)

The measurement behind this item is a size series on silicon, Cu K-alpha1 only, a 0.002 deg step, 100 000 counts at the strongest line and no counting noise. The conditions are chosen so that everything in Table 4 is the size and nothing else: one phase, one instrument, one reflection set, and enough counts that the statistical error is negligible against the widths being compared.

D (nm)	FWHM 111 (deg)	FWHM 220 (deg)	FWHM 400 (deg)	height 111 (counts)	area 111 (counts deg)	Scherrer 111 (nm)
5	1.6334	1.7279	1.8931	96	148	5
10	0.8204	0.8680	0.9601	192	146	10
20	0.4130	0.4368	0.4852	382	160	20
50	0.1721	0.1816	0.2018	916	162	50
100	0.0973	0.1020	0.1133	1620	162	100
200	0.0666	0.0691	0.0765	2368	162	200
500	0.0550	0.0565	0.0624	2869	162	499

Table 4: the size study. The Scherrer column subtracts the instrument width in quadrature and applies K lambda/(D cos theta) with K = 0.9 [3]. The area column integrates over a window of four FWHM taken from the scan range, so for the broadest peak (5 nm) it retains only part of the tails: 148 against 162 counts deg for the large sizes is a windowing artefact, not a loss of power.

4.1 The peak position does not move

Silicon 111 sits at 28.441 deg for every size in Table 4, from 5 nm to 500 nm. Bragg's law fixes the angle from the cell, and the cell of a crystallite does not depend on how large the crystallite is. A size effect that shifted the lines would be a different measurement altogether: a strain or a composition change, not a size change.

4.2 The width grows as 1/D

The 111 width runs from 0.0550 deg at 500 nm to 1.633 deg at 5 nm, a factor of 30 for a hundredfold change in size. On log-log axes the low-angle points lie on a line of slope minus one (Figure 4, left), which is the signature of a width proportional to 1/D. The same slope appears for 220 and 400: the size effect is not a property of one reflection.

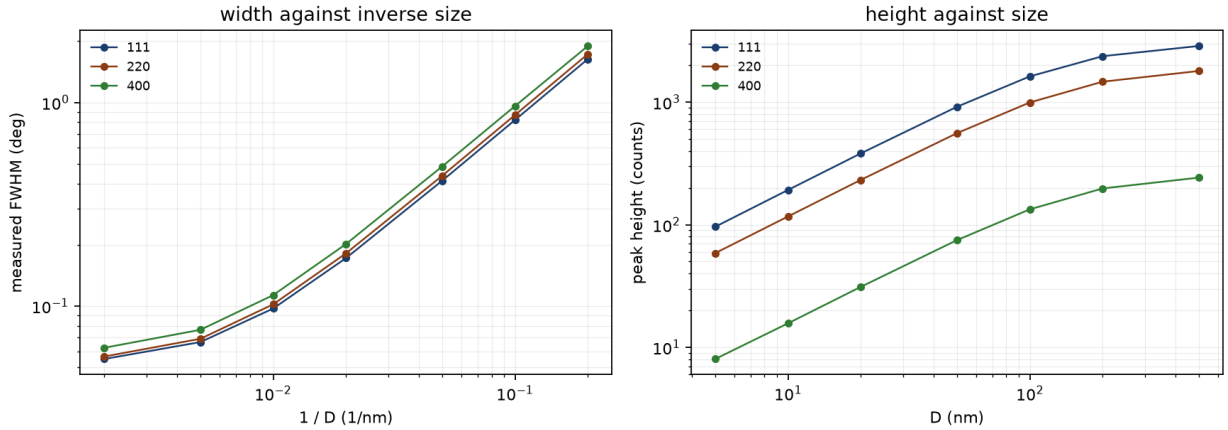


Figure 4: left, measured FWHM against 1/D on log-log axes, one line per reflection; right, peak height against D. Both families bend over where the instrument width takes control.

4.3 The height falls, the area does not

The peak height at 111 falls from 2869 counts at 500 nm to 96 counts at 5 nm, a factor of 30, close to the width factor. The integrated area, however, stays between 148 and 162 counts deg across the whole range: the diffracted power of that reflection is fixed by the structure factor and the beam, and a smaller crystallite spreads the same power over a wider angular range. This is the observation that section 5 explains in system terms, and the one that separates a size effect from an intensity effect.

4.4 The effect depends on the reflection

For one crystallite, high-angle lines are broader than low-angle ones: at 20 nm the 111 line is 0.413 deg and the 400 line is 0.485 deg, and Figure 5 shows the same trend across 20-120 deg. The cause is the 1/cos(theta) in the Scherrer relation; section 5.4 derives it as the change of variable between reciprocal space and the goniometer angle, so it is geometry, not a second physical effect.

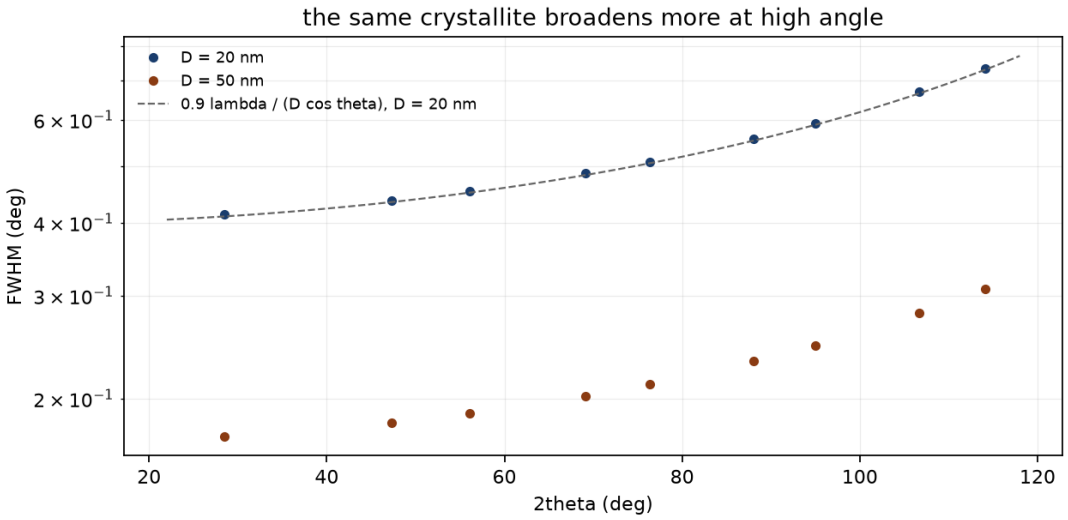


Figure 5: FWHM against 2theta for 20 nm and 50 nm crystallites, with K lambda/(D cos theta) drawn for 20 nm. The curve is not fitted.

4.5 The instrument sets a floor, and the recovery fails above it

At 500 nm the crystallite contribution is 0.016 deg, well below the instrument width of 0.0524 deg. What the scan measures is the instrument, and the Scherrer inversion still returns 499 nm only because the deconvolution is done on the same model that generated the data. On a real instrument the calibration would carry its own error, and above roughly 100 nm the size would be unmeasurable from the width alone. Below that the sample dominates and the inversion recovers the input size at every point of Table 4, from 5 nm to 500 nm.

The practical consequence is resolution. A 5 nm 111 line is 1.6 deg wide, wider than the separation between neighbouring reflections at low angle, so the lines merge into one envelope and a peak search that assumes separate maxima begins to fail. That is the same statement as a resolution limit in any measurement system, and it is why the analysis code in this repository carries a peak finder with an explicit height threshold rather than a simple maximum search.

4.6 How well does the width law hold?

Table 5 compares the measured width of every reflection in the study with the Scherrer prediction, K lambda/(D cos theta), using the same K and lambda the simulator uses. The residuals are largest where the quadrature combination of the two widths is least exact, that is at the smallest sizes and at the highest angle, and they are of the order of a few per cent of the width itself.

D (nm)	hkl	2theta (deg)	predicted (deg)	measured (deg)	residual (deg)	relative error
5	111	28.441	1.6391	1.6334	-0.0057	-0.3 %
5	220	47.301	1.7345	1.7279	-0.0067	-0.4 %
5	400	69.128	1.9294	1.8931	-0.0363	-1.9 %
10	111	28.441	0.8195	0.8204	+0.0008	+0.1 %
10	220	47.301	0.8673	0.8680	+0.0007	+0.1 %
10	400	69.127	0.9647	0.9601	-0.0046	-0.5 %
20	111	28.441	0.4098	0.4130	+0.0032	+0.8 %
20	220	47.301	0.4336	0.4368	+0.0032	+0.7 %
20	400	69.127	0.4823	0.4852	+0.0028	+0.6 %

D (nm)	hkl	2theta (deg)	predicted (deg)	measured (deg)	residual (deg)	relative error
50	111	28.441	0.1639	0.1721	+0.0082	+5.0 %
50	220	47.301	0.1735	0.1816	+0.0081	+4.7 %
50	400	69.127	0.1929	0.2018	+0.0089	+4.6 %
100	111	28.441	0.0820	0.0973	+0.0153	+18.7 %
100	220	47.301	0.0867	0.1020	+0.0153	+17.7 %
100	400	69.127	0.0965	0.1133	+0.0168	+17.4 %
200	111	28.441	0.0410	0.0666	+0.0256	+62.4 %
200	220	47.301	0.0434	0.0691	+0.0257	+59.3 %
200	400	69.127	0.0482	0.0765	+0.0282	+58.6 %
500	111	28.441	0.0164	0.0550	+0.0386	+235.3 %
500	220	47.301	0.0173	0.0565	+0.0392	+225.8 %
500	400	69.127	0.0193	0.0624	+0.0431	+223.4 %

Table 5: measured width against the Scherrer prediction. The measured column is the FWHM the analysis code extracts from the simulated scan; the predicted column is the algebraic size width that was put into it.

4.7 A worked example: reading a size from one scan

Table 7 takes the 20 nm row of Table 4 and applies the analysis the way it would be applied to a measured scan. For each reflection: read the peak angle and the FWHM at half height, subtract the instrument width in quadrature, and apply D = K lambda/(beta cos theta) with K = 0.9. The three reflections return the same size to better than 0.5 per cent, which is the internal consistency check that a size measurement is expected to pass.

Reflection	2theta (deg)	FWHM (deg)	instrument (deg)	beta (deg)	D (nm)
111	28.441	0.4130	0.0524	0.4097	20.0
220	47.301	0.4368	0.0538	0.4335	20.0
400	69.127	0.4852	0.0593	0.4816	20.0

Table 7: the inversion, reflection by reflection, for the 20 nm crystallite. The uncorrected widths differ by 17 per cent between 111 and 400; after the 1/cos(theta) correction the three sizes agree.

The example also shows why the correction matters. The raw widths increase from 0.413 deg at 111 to 0.485 deg at 400, an apparent 17 per cent difference that has nothing to do with the specimen; it is the geometry of equation (5). A routine that used the raw width would report a size that drifts with the reflection it happened to fit, and a report that quoted one reflection without saying which would not be reproducible.

5. Item 3: why it happens, in signals-and-systems terms (25%)

5.1 The pattern is a Fourier transform

The amplitude scattered by a specimen is the Fourier transform of its electron density, A(q) proportional to the integral of rho(r) exp(-i q.r) dr, with q = 4 pi sin(theta)/lambda [2, ch. 2]. An infinite crystal is a lattice convolved with a unit-cell motif, so its transform is a reciprocal lattice multiplied by the structure factor: delta functions at the Bragg points. That is the top panel of Figure 6, a line with no width.

A real crystal is that lattice multiplied by a window w(r), equal to one inside the crystallite and zero outside. Multiplication in real space is convolution in reciprocal space [4, ch. 3], [5, ch. 2], so the amplitude becomes

```
A(q) = [reciprocal lattice x F(q)] * W(q),    W = FT{w},
```

and the measured intensity carries |W|^2. The finite size of the crystal is therefore not a correction applied to the physics; it is a filter applied to the ideal spectrum. Everything that follows is a property of that filter.

5.2 The filter is the transform of a rectangle

Take a crystal of N planes with spacing d, so D = N d, and write the window as a rectangle of that length. The intensity near a Bragg point is the squared magnitude of the finite geometric sum, the interference function

```
I(q) proportional to |sin(N q d / 2) / sin(q d / 2)|^2,    (2)
```

which reduces to a sinc-squared envelope, width proportional to 1/N, as soon as q is small enough that sin(qd/2) is close to qd/2. Its first zeros sit at q = 2 pi / D, and its full width at half maximum is a fixed multiple of that scale: the numerical value is 0.886 x 2 pi / D for large N, 0.890 for N = 10 (section 5.3). That number is the shape constant K of the Scherrer relation. The exact value for a finite chain is 0.8859, and the conventional 0.9 is it rounded. K is not a constant of nature: Scherrer's own 1918 paper gives 0.94 for a cube [10], the value depends on the shape of the crystallite and on which reflection is used [7], and it differs again if the width is defined as an integral breadth instead of a half width [8]. Using K = 0.9 against the exact 0.886 therefore returns a size 1.6 per cent too large, which is well inside the scatter between the conventions.

The same result is available without writing the sum: the transform of a rectangle of length D is a sinc whose first zeros are at q = 2 pi / D, and the squared magnitude gives the observed profile [5]. The two routes must agree, and they do to the numerical values in section 5.3.

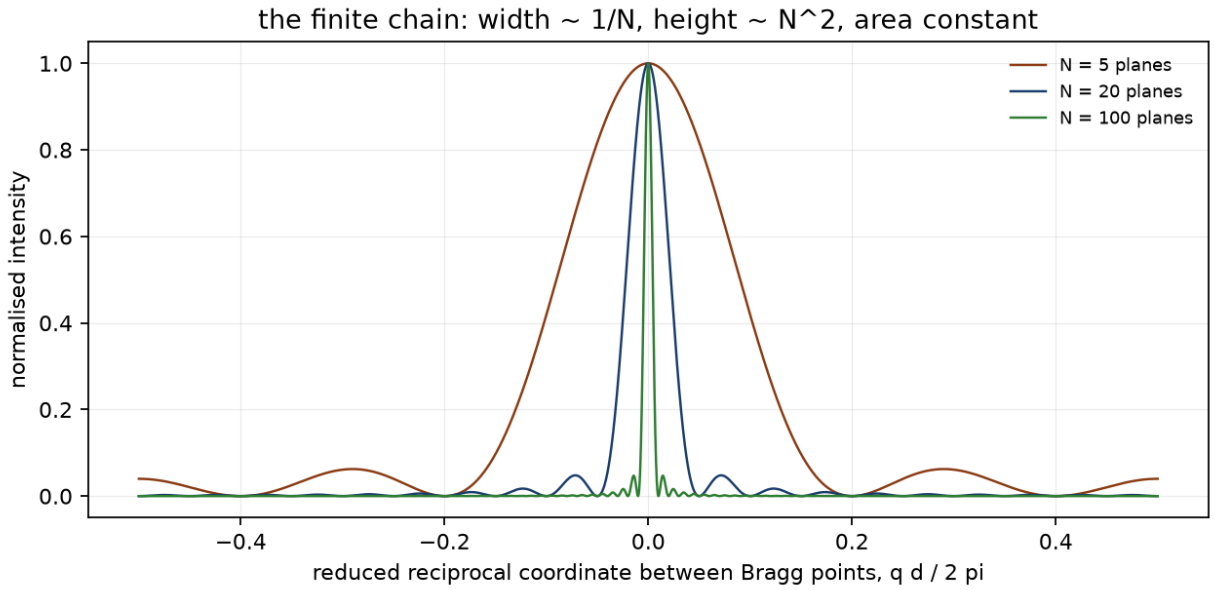


Figure 7: the interference function of equation (2) for N = 5, 20 and 100 planes, plotted against the reduced reciprocal coordinate q d / 2 pi, that is the distance between Bragg points measured in reciprocal-lattice units. The width falls as 1/N while the peak height rises as N^2, so the area under the line is unchanged.

5.3 The width constant, measured

Equation (2) can be evaluated directly. The table below reports the FWHM of the interference function in reciprocal space, normalised as FWHM x D / (2 pi), for several chain lengths:

N planes	FWHM x D / (2 pi)	implied K
10	0.8897	0.8897
20	0.8869	0.8869
50	0.8860	0.8860
100	0.8859	0.8859
1000	0.8859	0.8859

Table 6: the shape constant of a pure finite-chain profile, computed from equation (2). The Scherrer constant K = 0.9 used in the code is this value rounded up; a Gaussian or Lorentzian profile would give a slightly different number, which is why K is quoted as a convention rather than a constant of nature [1].

5.4 Two linear stages, and what follows from that

The chain is two linear, shift-invariant stages in series. The input is the Bragg impulse train; the crystal has impulse response h_crystal = W; the instrument has impulse response h_instrument. Convolution commutes and associates, so the observed line is the ideal line blurred by both, in either order [4, ch. 3]:

```
I_observed = I_ideal * h_crystal * h_instrument.    (3)
```

Bandwidth. A long impulse response has a narrow frequency response. A crystallite of N planes is a window of length N d, and the width of its transform scales as 1/N. Halving the crystallite doubles the width in q, which is the line of slope minus one in Figure 4. The reciprocal relation is the same statement as the time-bandwidth product of signals and systems: a short observation cannot resolve fine structure in frequency, and a small crystal cannot produce a sharp line.

Area. A normalised impulse response has unit area, and convolution preserves the integral. The area under a reflection is therefore fixed by the structure factor and the beam, while its height is area divided by width. Table 4 shows exactly that: the area at 111 stays within about ten per cent across a hundredfold change in size, while the height follows the width. In the language of the same theory this is Parseval's theorem: the total power in the line is unchanged, only its distribution in q is.

The cascade is limited by its widest stage. The instrument response is set by the slits, the source size and the detector, and for any crystallite above roughly 100 nm it is wider than the crystal response. That is why the curves in Figure 4 flatten at the right, and why the recovery in Table 4 stops being meaningful there: the measurement can no longer see the narrower of the two filters. Adding two Gaussian widths in quadrature is the statement that variances add when independent responses are convolved; the Lorentzian part of a pseudo-Voigt adds linearly instead, which is why the quadrature correction in Table 4 is approximate and why the residuals in Table 5 grow at the smallest sizes.

5.5 The 1/cos(theta) is a change of variable

The broadening happens in q, but the scan is recorded in 2theta. The two are related by q = 4 pi sin(theta)/lambda, so

```
dq/d(2theta) = (2 pi / lambda) cos(theta),    (4)
```

and a fixed width in q appears in the scan as

```
Delta(2theta) = Delta(q) / |dq/d(2theta)| = K lambda / (D cos theta).    (5)
```

The 1/cos(theta) is a Jacobian: the change of variable between the frequency axis of the spectrum and the goniometer angle on which it is displayed. At high angle a degree of 2theta spans fewer reciprocal-space units, so the same spectral width looks wider on the chart. The same variable change explains why the K-alpha doublet separates more at high angle, which is visible in Figure 5 of the companion doublet figure in the repository. This is why a size measured from one low-angle line and one high-angle line should agree once equation (5) is applied, and Table 5 is that check.

5.6 Sampling, and why the scan step matters

The detector samples the profile at the step size of the scan. A peak measured with too few points per width is undersampled in exactly the sense of the sampling theorem: its height and its width become functions of where the samples happen to fall, and a fit to three points does not recover the continuous line. In the study the narrowest line is the 500 nm case, whose width is about 0.055 deg; a 0.02 deg step gives under three points per FWHM, and the measured width then depends on the step. The 0.002 deg step used for Table 4 is chosen to sample the narrowest line by about 27 points, which is why the numbers in that table can be compared with each other.

This is the same trade-off as bandwidth against acquisition time in any measurement: sampling finer costs counts per point or time, and sampling coarser throws away the information the width carries. The size of the effect is measurable: at 500 nm the same scan taken at a 0.02 deg step reports a width 4.2 per cent larger and a size 28.5 per cent smaller than the 0.002 deg scan, while at 0.002 deg the error is 0.3 per cent. A real experiment that wants a size below 10 nm must choose its step from the expected width, not from habit.

5.7 What the model predicts, and what it measures

None of section 5 is fitted. The curve in Figure 5 is equation (5) with K = 0.9 and the simulator's own wavelength; the widths in Table 5 come from the simulated scans. At 20 nm the prediction for 111 is 0.410 deg against 0.413 measured; at 400, 0.482 against 0.485; at 5 nm, 1.64 against 1.63. The agreement is the evidence that the model's size term is the convolution argument above and not an empirical fit, and the residuals in Table 5 say where the approximation (quadrature combination

of a pseudo-Voigt) is weakest.

Two other descriptions of the same physics are common and are worth naming, because a reader will meet them. The Warren-Averbach analysis works with the Fourier coefficients of the line profile directly, which is the most literal form of section 5.1 and separates size from strain by their different order dependence [2]. It exists because a single breadth cannot do that on its own: the breadth gives a mean dimension, and the shape of the size distribution appears only in the higher-order coefficients [9]. The Williamson-Hall plot uses several reflections to separate the 1/cos and tan terms; the analysis code in this repository implements it as a check and returns the input size and zero strain on a size-only simulation. Neither is needed for item 3, and neither is used for the numbers in this report, but both are the same convolution argument written in a different basis.

6. Verification, uncertainty and limits

Every number in this report is produced by the simulator that draws the figures, and the report is gated on that. Peak positions are checked against the NIST SRM 640f certificate, every phase against its published strongest line, and the whole document rebuilds byte-identically from a fresh clone of the repository. The test suite covers the lattice geometry, the structure factor and its absences, the profile, the detector and the scan, and it includes an independent brute-force count of the symmetry orbits and a metric-tensor check of the d-spacings.

The uncertainty in the size numbers has four parts. The counting noise is negligible here by construction: 100 000 counts at the peak give a statistical error below 0.4 per cent, far below the differences being compared. The profile shape matters less than one might expect: sweeping eta from 0 to 1 changes the peak height by about a third but the measured FWHM by less than 0.1 per cent at 20 nm, because the width is read at half maximum and the mixture is normalised to unit area. The quadrature combination of the instrumental and size widths is exact only for two Gaussians; with a pseudo-Voigt the Lorentzian part adds linearly, and that is the main reason the residuals in Table 5 grow at the smallest sizes. Finally the instrument constants U, V and W are plausible defaults rather than a calibration of a real diffractometer, so the absolute widths belong to this model; the scaling laws, which are what items 2 and 3 are about, do not depend on them.

What could change the conclusions? A different shape constant would move every recovered size by the same factor and would not change the 1/D law. A different instrument width would move the crossover between the sample-limited and instrument-limited regimes, but not the slope. A specimen with strain would broaden the lines with a tan(theta) term, which the size-only analysis would misread as a smaller crystallite at high angle; that is precisely what the Williamson-Hall check in the repository is for, and why this report uses a strain-free simulation.

Deliberately not modelled: Rietveld refinement, anomalous dispersion, preferred orientation, K-beta residual, detector energy discrimination, thin-film absorption, axial-divergence asymmetry, amorphous background and absolute intensities. Every site shares one Debye-Waller coefficient. The Cu K-alpha wavelengths are the conventional values; no primary table was read for them. None of these affects the size argument, which rests on the width and on the convolution.

6.1 Uncertainty budget

The table collects the terms discussed above, each measured on this model rather than estimated. The two that matter for a size measurement are the scan step on a narrow line and the instrument width itself; the others are below the differences the report is asked to demonstrate.

Source	Effect measured	Where it matters
Counting noise	below 0.4 % on the peak height at 100 000 counts	negligible here by construction
Profile shape, eta 0 to 1	width below 0.1 %, height about 32 %	height comparisons, not widths
Scan step, 0.02 deg at 500 nm	width +4.2 %, recovered size -28.5 %	narrow lines, large crystallites
Quadrature subtraction	up to 1.9 % on the width at 5 nm, 400	the smallest sizes
Instrument U, V, W	sets the crossover near 100 nm	any size above about 100 nm
Wavelength convention	0.0001 deg at 30 deg	negligible
Cell constants	below 0.02 deg in 2theta	positions, not widths

Table 9: the uncertainty budget. The step and eta entries were measured by sweeping the parameter and re-running the analysis; the quadrature entry is the largest residual in Table 5.

7. Conclusion

Item 1: the simulator described in section 2 reproduces the diffraction patterns of nine crystal structures and of one structure at six crystallite sizes (Figures 2 and 3), and its code is Appendix A. Item 2: changing the domain size changes the width of every line as 1/D, changes the height in proportion, leaves the integrated area and the Bragg positions alone, and does so more strongly at high angle. Item 3: those four facts are the properties of one filter. A finite crystallite multiplies the lattice by a window, convolution in reciprocal space turns that window into a sinc-squared of width 2 pi / D, the instrument is a second convolution in series, and the 1/cos(theta) is the Jacobian between reciprocal space and the goniometer angle. The measurements in Table 4 and Table 5 agree with that description without fitting anything.

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Every entry was checked on 2026-10-22: each DOI was resolved through Crossref and each ISBN through OpenLibrary, and each work is cited for the claim it supports. The checks are recorded in report/references/AUDIT.md.

Appendix A: the code

Every module the simulator ships, in alphabetical order. The same files are in the repository under assignments/02-xrd-diffractometer/xrdsim/.

xrdsim/__init__.py

```
"""XRD diffractometer simulator (assignment 2, quantum physics section)."""
__version__ = "0.1.0"
```

xrdsim/__main__.py

```
from .cli import main

raise SystemExit(main())
```

xrdsim/analysis.py

```
"""Reading a simulated or measured pattern: peak finding, widths and indexing.

The functions here work on arrays (2theta, counts) and make no assumption about where
they came from, so the same code is used on simulator output and on a measured scan.
"""

from __future__ import annotations
import math
```

```
from dataclasses import dataclass

import numpy as np

from .profile import scherrer_size


@dataclass(frozen=True)
class Peak:
    two_theta: float
    height: float
    fwhm: float
    area: float


def find_peaks(two_theta: np.ndarray, counts: np.ndarray,
               min_height_fraction: float = 0.02) -> list[Peak]:
    """Local maxima above a fraction of the tallest point, with width and area."""
    x = np.asarray(two_theta, dtype=float)
    y = np.asarray(counts, dtype=float)
    if x.size != y.size or x.size < 3:
        raise ValueError("need matching arrays of at least three points")
    threshold = min_height_fraction * y.max()
    peaks: list[Peak] = []
    for i in range(1, len(y) - 1):
        if not (y[i] >= y[i - 1] and y[i] > y[i + 1] and y[i] >= threshold):
            continue
        centre, height = _refine(x, y, i)
        half = (height + _local_floor(x, y, i)) / 2.0
        fwhm = _width_at(x, y, i, half)
        area = _area(x, y, i, 4.0 * fwhm if fwhm > 0 else x[1] - x[0])
        peaks.append(Peak(centre, height, fwhm, area))
    return peaks


def _local_floor(x: np.ndarray, y: np.ndarray, i: int, reach_deg: float = 10.0) -> float:
    """Baseline near a peak: the 10th percentile within `reach_deg`, so a neighbouring
    reflection cannot drag the half-height down and the result does not depend on the step."""
    step = x[1] - x[0] if len(x) > 1 else 1.0
    reach = max(1, int(reach_deg / step))
    left = max(0, i - reach)
    right = min(len(y), i + reach)
    return float(np.percentile(y[left:right], 10))


def _width_at(x: np.ndarray, y: np.ndarray, i: int, half: float) -> float:
    left = i
    while left > 0 and y[left] > half:
        left -= 1
    right = i
    while right < len(y) - 1 and y[right] > half:
        right += 1
    if left == i or right == i:
        return 0.0
    x_left = np.interp(half, [y[left], y[left + 1]], [x[left], x[left + 1]])
    x_right = np.interp(half, [y[right], y[right - 1]], [x[right], x[right - 1]])
    return float(x_right - x_left)


def _area(x: np.ndarray, y: np.ndarray, i: int, half_window: float) -> float:
    window = np.abs(x - x[i]) <= half_window
    return float(np.trapezoid(y[window] - np.median(y[window]), x[window]))


def pure_width(observed: float, instrumental: float) -> float:
    """Sample-only FWHM by Gaussian quadrature subtraction, guarded at zero."""
    value = observed * observed - instrumental * instrumental
    return math.sqrt(value) if value > 0 else 0.0


def scherrer_from_peak(peak: Peak, instrumental_fwhm: float, wavelength: float,
                       k: float = 0.9) -> float:
    """Crystallite size in angstrom from one peak, corrected for instrumental width."""
    beta = pure_width(peak.fwhm, instrumental_fwhm)
    if beta <= 0:
        raise ValueError("the peak is not broader than the instrument")
    return scherrer_size(beta, peak.two_theta, wavelength, k)


def index_cubic(two_thetas: list[float],
                max_n_squared: int = 30, tolerance: float = 0.02) -> list[tuple[tuple[int, int, int], float]]:
    """Assign (hkl) to a cubic pattern by the sin^2(theta) ratios.

    The cell returned is the smallest one that fits, so a body-centred pattern comes back
```

```
    as a primitive cell with half the conventional edge; `lattice_hint` names that.

    Returns the assignments and the implied n^2 = h^2 + k^2 + l^2. The first reflection
    is taken as the lowest-angle one; every other angle must fall on an integer multiple
    of its sin^2(theta) to within `tolerance` (relative).
    """
    if not two_thetas:
        raise ValueError("no peaks to index")
    two_thetas = sorted(two_thetas)
    s2 = [math.sin(math.radians(t / 2.0)) ** 2 for t in two_thetas]
    base = s2[0]
    for n2_first in range(1, max_n_squared + 1):
        ratios = [value / base * n2_first for value in s2]
        if all(abs(r - round(r)) <= tolerance * max(round(r), 1) for r in ratios):
            assignments = []
            for angle, ratio in zip(two_thetas, ratios):
                n2 = int(round(ratio))
                assignments.append((_hkl_from_n2(n2), angle))
            return assignments
    raise ValueError("the pattern does not index as cubic")


def lattice_hint(n_squared: list[int]) -> str:
    """Describe the lattice from the observed n^2 sequence.

    All-even n^2 means a body-centred lattice described on half the cell; the FCC sequence
    starts 3, 4, 8, 11, 12, 16, 19, 20, 24. Anything else is primitive.
    """
    values = sorted({int(value) for value in n_squared})
    fcc = {3, 4, 8, 11, 12, 16, 19, 20, 24, 27, 32, 35, 36, 40, 43, 44, 48}
    if values and all(value % 2 == 0 for value in values):
        return "body-centred, conventional cell twice the refined edge"
    if set(values) <= fcc:
        return "face-centred (the 3, 4, 8, 11 sequence)"
    if 1 in values:
        return "primitive cell; a body-centred sample would have twice this edge"
    return "primitive lattice"


def _hkl_from_n2(n2: int) -> tuple[int, int, int]:
    for h in range(0, 8):
        for k in range(0, 8):
            l_sq = n2 - h * h - k * k
            if l_sq < 0:
                continue
            l = round(math.isqrt(l_sq))
            if l * l == l_sq:
                return tuple(sorted((h, k, l), reverse=True))
    raise ValueError(f"no (hkl) has n^2 = {n2}")


def _refine(x: np.ndarray, y: np.ndarray, i: int) -> tuple[float, float]:
    """Parabolic refinement of a sampled maximum: position and height between bins."""
    y0, y1, y2 = y[i - 1], y[i], y[i + 1]
    denom = y0 - 2.0 * y1 + y2
    if denom == 0:
        return float(x[i]), float(y[i])
    delta = 0.5 * (y0 - y2) / denom
    if abs(delta) > 0.5:
        return float(x[i]), float(y[i])
    return float(x[i] + delta * (x[i + 1] - x[i])), float(y1 - 0.25 * (y0 - y2) * delta)


def williamson_hall(peaks: list[Peak], instrumental_fwhm_for,
                    wavelength: float, k: float = 0.9) -> tuple[float, float]:
    """Crystallite size and microstrain from the Williamson-Hall plot.

    beta cos(theta) = K lambda / D + 4 epsilon sin(theta), fitted by least squares over the
    peaks given. `instrumental_fwhm_for(two_theta)` returns the instrument width at that
    angle. Returns (size in angstrom, strain) and raises when fewer than two peaks are
    usable.
    """
    xs, ys = [], []
    for peak in peaks:
        beta = pure_width(peak.fwhm, instrumental_fwhm_for(peak.two_theta))
        if beta <= 0 or peak.two_theta <= 0:
            continue
        theta = math.radians(peak.two_theta / 2.0)
        xs.append(4.0 * math.sin(theta))
        ys.append(math.radians(beta) * math.cos(theta))
    if len(xs) < 2:
        raise ValueError("Williamson-Hall needs at least two peaks broader than the instrument")
    slope, intercept = np.polyfit(xs, ys, 1)
    if intercept <= 0:
```

```
        raise ValueError("the fit has no positive intercept, so no size can be reported")
    return k * wavelength / intercept, float(slope)


def identify_phase(two_thetas: list[float], phase, instrument,
                   tolerance: float = 0.15,
                   min_intensity_fraction: float = 0.15) -> float:
    """Fraction of a phase's strong reflections that the observed peaks contain.

    A score of 1.0 means every predicted line above `min_intensity_fraction` of the
    phase's strongest has an observed peak within `tolerance` degrees.
    """
    from .diffractometer import reflections
    rows = reflections(phase, instrument)
    if not rows:
        return 0.0
    strongest = max(row.intensity for row in rows)
    predicted = [row.two_theta for row in rows
                 if row.intensity > min_intensity_fraction * strongest]
    if not predicted or not two_thetas:
        return 0.0
    matched = sum(1 for angle in predicted
                  if min(abs(angle - seen) for seen in two_thetas) <= tolerance)
    return matched / len(predicted)


def identify(two_thetas: list[float], phases: dict, instrument,
             tolerance: float = 0.15) -> list[tuple[str, float]]:
    """Rank the phases of a database by how well they explain the observed peak positions."""
    scored = [(key, identify_phase(two_thetas, phase, instrument, tolerance))
              for key, phase in phases.items()]
    return sorted(scored, key=lambda row: (-row[1], row[0]))
```

xrdsim/atoms.py

```
"""Atomic scattering factors and the Debye-Waller term.

`f0` is interpolated from data/scattering_factors.json, sampled once from the
Waasmaier-Kirfel analytic form factors (Acta Cryst. A51, 416-431, 1995).
`s = sin(theta) / lambda` in 1/angstrom.
"""

from __future__ import annotations

import json
import math
import pathlib

DATA_PATH = pathlib.Path(__file__).resolve().parent.parent / "data" / "scattering_factors.json"

_table: dict | None = None


def _load() -> dict:
    global _table
    if _table is None:
        payload = json.loads(DATA_PATH.read_text(encoding="utf-8"))
        _table = {
            "s_max": payload["grid_s_inv_A"][-1],
            "step": payload["grid_s_inv_A"][1] - payload["grid_s_inv_A"][0],
            "elements": payload["f0"],
        }
    return _table


def f0(element: str, s: float) -> float:
    """Elastic form factor f0 in electrons. `s = sin(theta)/lambda` in 1/angstrom."""
    table = _load()
    values = table["elements"].get(element)
    if values is None:
        raise KeyError(f"no scattering factor for {element!r}; rebuild data/scattering_factors.json")
    if s < 0:
        raise ValueError("s must be non-negative")
    position = s / table["step"]
    last = len(values) - 1
    if position >= last:
        return float(values[last])
    i = int(position)
    frac = position - i
    return float(values[i] * (1.0 - frac) + values[i + 1] * frac)


def debye_waller(b_iso: float, s: float) -> float:
```

```
    """Temperature factor exp(-B s^2), with B = 8 pi^2 <u^2> in square angstrom."""
    if b_iso < 0:
        raise ValueError("B must be non-negative")
    return math.exp(-b_iso * s * s)
```

xrdsim/cli.py

```
"""Command line interface: `python -m xrdsim`."""

from __future__ import annotations

import argparse
import pathlib

from . import sources
from .analysis import (find_peaks, identify, index_cubic, lattice_hint,
                       scherrer_from_peak, williamson_hall)
from .diffractometer import Instrument, reflections, scan
from .profile import caglioti_fwhm
from .samples import PHASES, get_phase


def _instrument(args) -> Instrument:
    return Instrument(
        ka2=not args.no_ka2,
        monochromator_two_theta=args.monochromator,
        reference_counts=args.counts,
        background_rate=args.background,
        dwell_s=args.dwell,
        rng_seed=None if args.noise == 0 else args.noise,
        sample_displacement_mm=args.displacement,
    )


def main(argv: list[str] | None = None) -> int:
    parser = argparse.ArgumentParser(prog="xrdsim", description=__doc__)
    parser.add_argument("--phase", help="phase key (see --list-phases)")
    parser.add_argument("--range", nargs=2, type=float, metavar=("MIN", "MAX"),
                        default=[20.0, 90.0], help="2theta scan range in degrees")
    parser.add_argument("--step", type=float, default=0.02, help="2theta step in degrees")
    parser.add_argument("--dwell", type=float, default=1.0, help="counting time per step, s")
    parser.add_argument("--counts", type=float, default=20_000.0,
                        help="integrated counts at the strongest reflection")
    parser.add_argument("--background", type=float, default=0.5, help="background, counts/s")
    parser.add_argument("--displacement", type=float, default=0.0, metavar="MM",
                        help="sample surface displaced from the goniometer axis")
    parser.add_argument("--size", type=float, default=None,
                        help="crystallite size in angstrom (adds size broadening)")
    parser.add_argument("--no-ka2", action="store_true", help="drop the K-alpha2 line")
    parser.add_argument("--monochromator", type=float, default=None,
                        help="monochromator 2theta in degrees")
    parser.add_argument("--noise", type=int, default=0,
                        help="Poisson seed; 0 disables counting noise")
    parser.add_argument("--out", type=pathlib.Path, help="write the scan to this image")
    parser.add_argument("--table", action="store_true", help="print the reflection table")
    parser.add_argument("--index", action="store_true",
                        help="index the observed peaks as a cubic pattern")
    parser.add_argument("--scherrer", action="store_true",
                        help="report Scherrer sizes for the observed peaks")
    parser.add_argument("--identify", action="store_true",
                        help="rank the phase database against the simulated pattern")
    parser.add_argument("--list-phases", action="store_true", help="list the phase database")
    args = parser.parse_args(argv)

    if args.list_phases:
        for key, phase in sorted(PHASES.items()):
            print(f"{key:14s} {phase.formula:8s} {phase.space_group:18s} {phase.source}")
        return 0
    if not args.phase:
        parser.error("--phase is required unless --list-phases is given")

    try:
        return _run(args)
    except (KeyError, ValueError) as error:
        parser.error(str(error).strip('"'))


def _run(args) -> int:
        phase = get_phase(args.phase)
        instrument = _instrument(args)
        result = scan(phase, args.range[0], args.range[1], args.step, instrument,
                      size_a=args.size)
        if args.table:
```

```
            strongest = max(row.intensity for row in result.reflections)
            print(f"{'hkl':>10s} {'d (A)':>8s} {'2theta':>8s} {'m':>3s} {'I/I0':>7s}")
            for row in result.reflections:
                if row.intensity < 1e-4 * strongest:
                    continue
                label = "".join(str(v) for v in row.hkl)
                print(f"{label:>10s} {row.d:8.4f} {row.two_theta:8.3f} {row.multiplicity:3d} "
                      f"{100 * row.intensity / strongest:7.2f}")
            return 0

        peaks = find_peaks(result.two_theta, result.counts)
        print(f"phase      : {phase.name} ({phase.formula}), {phase.space_group}")
        print(f"wavelength : {instrument.wavelength:.6f} A"
              + (" + K-alpha2" if instrument.ka2 else ""))
        print(f"range      : {args.range[0]} - {args.range[1]} deg, step {args.step}")
        print(f"peaks      : {len(peaks)}")
        for peak in peaks[:12]:
            line = f"  {peak.two_theta:8.3f} deg  height {peak.height:9.0f}  FWHM {peak.fwhm:.4f}"
            if args.size and not args.scherrer:
                try:
                    size = scherrer_from_peak(peak, caglioti_fwhm(peak.two_theta, *instrument.caglioti),
                                              instrument.wavelength)
                    line += f"  Scherrer {size / 10.0:.1f} nm"
                except ValueError:
                    pass
            print(line)

        if args.index:
            try:
                rows = index_cubic([peak.two_theta for peak in peaks])
            except ValueError as error:
                print(f"indexing failed: {error}")
            else:
                print("indexing   : hkl            2theta")
                for hkl, angle in rows:
                    print(f"             {str(hkl):14s} {angle:7.3f}")
                n_squared = [hkl[0] ** 2 + hkl[1] ** 2 + hkl[2] ** 2 for hkl, _ in rows]
                print(f"lattice    : {lattice_hint(n_squared)}")

        if args.scherrer:
            print("Scherrer   : peak            size (nm)")
            for peak in peaks:
                try:
                    size = scherrer_from_peak(
                        peak, caglioti_fwhm(peak.two_theta, *instrument.caglioti),
                        instrument.wavelength)
                except ValueError:
                    continue
                print(f"             {peak.two_theta:7.3f}         {size / 10.0:6.1f}")
            try:
                wh_size, wh_strain = williamson_hall(
                    peaks, lambda angle: caglioti_fwhm(angle, *instrument.caglioti),
                    instrument.wavelength)
            except ValueError as error:
                print(f"Williamson-Hall: {error}")
            else:
                print(f"Williamson-Hall: D = {wh_size / 10.0:.1f} nm, strain = {wh_strain:.5f}"
                      + ("" if args.no_ka2 else " (use --no-ka2: the shoulders bias the fit)"))

        if args.identify:
            ranking = identify([peak.two_theta for peak in peaks], PHASES, instrument)
            print("identification: phase          score")
            for key, score in ranking[:5]:
                print(f"               {key:14s} {score:5.2f}")

        if args.out:
            from .plot import save_scan
            args.out.parent.mkdir(parents=True, exist_ok=True)
            save_scan(result, args.out,
                      title=f"{phase.name}, {instrument.wavelength:.4f} A")
            print(f"wrote {args.out}")
        return 0
```

xrdsim/detector.py

```
"""Detector: counting statistics and dead time.

A powder scan is a counting experiment. The model returns expected counts per step; the
detector turns them into observed counts with Poisson noise.
"""

from __future__ import annotations
import math
```

```
def expected_counts(intensity: float, dwell_s: float) -> float:
    """Mean counts at a step: intensity (counts per second) times dwell time."""
    if dwell_s <= 0:
        raise ValueError("dwell time must be positive")
    return max(intensity, 0.0) * dwell_s


def observe(mean_counts: float, rng) -> float:
    """One Poisson draw for a step. `rng` is a numpy Generator."""
    return float(rng.poisson(max(mean_counts, 0.0)))


def count_noise(mean_counts: float) -> float:
    """Standard deviation of a Poisson count (sqrt(N)); zero for an empty step."""
    return math.sqrt(max(mean_counts, 0.0))


def apply_dead_time(true_rate: float, tau_s: float) -> float:
    """Observed rate for a non-paralysable detector: n = m / (1 + m tau)."""
    if true_rate < 0 or tau_s < 0:
        raise ValueError("rates and dead time must be non-negative")
    return true_rate / (1.0 + true_rate * tau_s)
```

xrdsim/diffractometer.py

```
"""The diffractometer: reflections, peak profiles and the scan itself.

A scan is built in counts per step. Reflection intensities are relative (I/I0), as in
Assignment 1: the model reproduces patterns and intensity ratios, not absolute
radiometry.
"""

from __future__ import annotations

import math
from dataclasses import dataclass, field

import numpy as np

from . import sources
from .detector import apply_dead_time, expected_counts
from .lattice import d_spacing
from .profile import (add_in_quadrature, caglioti_fwhm, pseudo_voigt_array,
                      size_fwhm)
from .samples import Phase
from .structure import lorentz_polarisation, multiplicity, structure_factor


@dataclass(frozen=True)
class Instrument:
    wavelength: float = sources.CU_KA1
    ka2: bool = True
    monochromator_two_theta: float | None = None
    caglioti: tuple[float, float, float] = (0.004, -0.002, 0.003)
    eta: float = 0.5
    reference_counts: float = 20_000.0     # counts at the strongest reflection
    background_rate: float = 0.5           # counts per second, flat
    dwell_s: float = 1.0
    dead_time_s: float = 0.0
    rng_seed: int | None = None      # None: no counting noise; an int seeds the Poisson draws
    max_two_theta: float = 150.0      # no goniometer reaches 180 deg, and LP diverges there
    sample_displacement_mm: float = 0.0  # specimen surface off the goniometer axis
    goniometer_radius_mm: float = 240.0


@dataclass(frozen=True)
class Reflection:
    hkl: tuple[int, int, int]
    d: float
    two_theta: float
    multiplicity: int
    f_squared: float
    intensity: float


@dataclass
class Scan:
    two_theta: np.ndarray
    counts: np.ndarray
    reflections: list[Reflection] = field(default_factory=list)
    phase: Phase | None = None
    instrument: Instrument | None = None
def reflections(phase: Phase, instrument: Instrument, hmax: int = 8) -> list[Reflection]:
```

```
    """Every reflection the wavelength reaches, with its relative intensity."""
    rows: list[Reflection] = []
    for h in range(0, hmax + 1):
        for k in range(0, hmax + 1):
            for l in range(0, hmax + 1):
                if h == 0 and k == 0 and l == 0:
                    continue
                if not _representative(h, k, l, phase.system):
                    continue
                try:
                    d = d_spacing(h, k, l, **phase.cell_kwargs())
                except ValueError:
                    continue
                if 2.0 * d < instrument.wavelength:
                    continue
                two_theta = 2.0 * math.degrees(
                    math.asin(instrument.wavelength / (2.0 * d)))
                if two_theta > instrument.max_two_theta:
                    continue
                f = structure_factor(phase, h, k, l, instrument.wavelength)
                m = multiplicity(h, k, l, phase.system)
                lp = lorentz_polarisation(two_theta, instrument.monochromator_two_theta)
                rows.append(Reflection((h, k, l), d, two_theta, m, abs(f) ** 2,
                                       abs(f) ** 2 * m * lp))
    rows.sort(key=lambda r: r.two_theta)
    return rows


def scan(phase: Phase, two_theta_min: float, two_theta_max: float, step: float,
         instrument: Instrument | None = None, size_a: float | None = None,
         hmax: int = 8) -> Scan:
    """Simulate one 2theta scan, with Poisson noise and optional crystallite size."""
    if instrument is None:
        instrument = Instrument()
    if step <= 0 or two_theta_max <= two_theta_min:
        raise ValueError("the scan range and step must be positive")
    grid = np.arange(two_theta_min, two_theta_max + step / 2.0, step)
    rows = reflections(phase, instrument, hmax)
    if not rows:
        raise ValueError("no reachable reflections in this range")

    strongest = max(row.intensity for row in rows)
    if strongest <= 0:
        raise ValueError("every reflection has zero intensity")

    lines = [(instrument.wavelength, 1.0)]
    if instrument.ka2:
        # the doublet is scaled from the instrument's own K-alpha1 so that another anode
        # keeps its own wavelengths instead of silently getting copper's
        ka2 = instrument.wavelength * sources.CU_KA2 / sources.CU_KA1
        lines = [(instrument.wavelength, 1.0), (ka2, sources.KA2_TO_KA1)]

    mean = np.full_like(grid, expected_counts(instrument.background_rate, instrument.dwell_s),
                        dtype=float)
    for row in rows:
        area_total = instrument.reference_counts * row.intensity / strongest
        for wavelength, weight in lines:
            d = d_spacing(*row.hkl, **phase.cell_kwargs())
            centre = 2.0 * math.degrees(math.asin(min(1.0, wavelength / (2.0 * d))))
            centre += displacement_shift(centre, instrument.sample_displacement_mm,
                                         instrument.goniometer_radius_mm)
            width = caglioti_fwhm(centre, *instrument.caglioti)
            if size_a is not None:
                width = add_in_quadrature(width, size_fwhm(size_a, centre, wavelength))
            if not (two_theta_min - 5.0 * width <= centre <= two_theta_max + 5.0 * width):
                continue
            profile = pseudo_voigt_array(grid, centre, width, instrument.eta)
            mean += area_total * weight * profile * step

    if instrument.dead_time_s:
        rate = mean / instrument.dwell_s
        mean = np.array([apply_dead_time(r, instrument.dead_time_s) * instrument.dwell_s
                         for r in rate])

    if instrument.rng_seed is None:
        counts = mean
    else:
        rng = np.random.default_rng(instrument.rng_seed)
        counts = rng.poisson(np.clip(mean, 0.0, None)).astype(float)

    return Scan(grid, counts, rows, phase, instrument)


def _representative(h: int, k: int, l: int, system: str) -> bool:
```

```
    """True for one member of each symmetry family.

    Enumerating h, k, l >= 0 visits every permutation of a reflection; the multiplicity
    already counts the whole orbit, so only the canonical member may be used.
    """
    if system == "cubic":
        return h >= k >= l
    if system == "tetragonal":
        return h >= k
    if system == "hexagonal":
        return h >= k
    return True


def displacement_shift(two_theta_deg: float, displacement_mm: float,
                       radius_mm: float) -> float:
    """2theta shift in degrees for a flat specimen displaced from the goniometer axis.

    Delta(2theta) = -2 s cos(theta) / R, in radians.
    """
    if displacement_mm == 0.0:
        return 0.0
    if radius_mm <= 0:
        raise ValueError('the goniometer radius must be positive')
    theta = math.radians(two_theta_deg / 2.0)
    return math.degrees(-2.0 * displacement_mm * math.cos(theta) / radius_mm)
```

xrdsim/geometry.py

```
"""Bragg-Brentano geometry: sample illumination and absorption.

Angles in degrees, lengths in mm, mass attenuation in cm^2/g. The flat-plate,
infinite-thickness case is the one a laboratory powder scan approximates.
"""

from __future__ import annotations

import math


def bragg_angle(two_theta_deg: float) -> float:
    """theta for a given 2theta."""
    return two_theta_deg / 2.0


def footprint_length(two_theta_deg: float, divergence_slit_mm: float,
                     sample_length_mm: float = 20.0) -> float:
    """Length of sample illuminated by the beam, in mm.

    The beam leaves the divergence slit with a fixed width, so the footprint on the
    flat plate grows as 1/sin(theta) and is clipped by the sample holder.
    """
    theta = math.radians(bragg_angle(two_theta_deg))
    if theta <= 0:
        raise ValueError("2theta must be positive")
    return min(sample_length_mm, divergence_slit_mm / math.sin(theta))


def linear_attenuation(density_g_cm3: float, mass_attenuation_cm2_g: float) -> float:
    """mu in 1/cm."""
    if density_g_cm3 <= 0 or mass_attenuation_cm2_g <= 0:
        raise ValueError("density and mass attenuation must be positive")
    return density_g_cm3 * mass_attenuation_cm2_g


def is_infinitely_thick(mu_per_cm: float, thickness_cm: float, threshold: float = 0.01) -> bool:
    """True when exp(-2 mu t / sin(theta)) is negligible; here the worst case theta -> 90 deg."""
    return math.exp(-2.0 * mu_per_cm * thickness_cm) < threshold


def flat_plate_absorption(mu_per_cm: float) -> float:
    """Absorption factor for a thick flat plate: A = 1 / (2 mu), independent of theta."""
    if mu_per_cm <= 0:
        raise ValueError("mu must be positive")
    return 1.0 / (2.0 * mu_per_cm)


def penetration_depth(mu_per_cm: float, two_theta_deg: float, fraction: float = 0.99) -> float:
    """Depth in cm that carries `fraction` of the diffracted intensity."""
    if not 0.0 < fraction < 1.0:
        raise ValueError("fraction must lie in (0, 1)")
    theta = math.radians(bragg_angle(two_theta_deg))
    return -math.log(1.0 - fraction) * math.sin(theta) / (2.0 * mu_per_cm)
```

xrdsim/lattice.py

```
"""Lattice geometry: d-spacings, Bragg angles and lattice-parameter refinement.

Lengths are in angstrom, angles in degrees. Only the settings a powder scan needs are
supported here: cubic, tetragonal, hexagonal and orthorhombic.
"""

from __future__ import annotations

import math


def d_spacing(h: int, k: int, l: int, *, a: float, b: float | None = None,
              c: float | None = None, hexagonal: bool = False) -> float:
    """Interplanar spacing in angstrom.

    `hexagonal=True` selects the hexagonal setting (a = b, gamma = 120 degrees).
    Otherwise `b` and `c` both None is cubic, `c` given with `b` None is tetragonal, and
    both `b` and `c` given is orthorhombic. A lone `b` is rejected rather than guessed.
    """
    if a <= 0:
        raise ValueError("lattice parameter must be positive")
    if hexagonal:
        if c is None:
            raise ValueError("the hexagonal setting needs c")
        inv_sq = 4.0 / 3.0 * (h * h + h * k + k * k) / a**2 + l * l / c**2
    elif b is None and c is None:
        inv_sq = (h * h + k * k + l * l) / a**2
    elif b is None:
        inv_sq = (h * h + k * k) / a**2 + l * l / c**2
    elif c is None:
        raise ValueError("an orthorhombic cell needs both b and c; for a tetragonal cell pass c and leave b No
    ne")
    else:
        inv_sq = h * h / a**2 + k * k / b**2 + l * l / c**2
    if inv_sq == 0:
        raise ValueError("(0,0,0) has no d-spacing")
    return 1.0 / math.sqrt(inv_sq)


def two_theta(d: float, wavelength: float) -> float:
    """Bragg angle in degrees for a d-spacing, or ValueError if the reflection is unreachable."""
    if d <= 0:
        raise ValueError("d-spacing must be positive")
    s = wavelength / (2.0 * d)
    if s > 1.0:
        raise ValueError(
            f"d = {d:.4f} A is unreachable at lambda = {wavelength:.4f} A (needs sin(theta) > 1)")
    return math.degrees(2.0 * math.asin(s))


def d_from_two_theta(two_theta_deg: float, wavelength: float) -> float:
    """d-spacing in angstrom from a measured 2theta, by Bragg's law."""
    theta = math.radians(two_theta_deg / 2.0)
    s = math.sin(theta)
    if s == 0:
        raise ValueError("2theta = 0 carries no information")
    return wavelength / (2.0 * s)


def refine_cubic_a(peaks: list[tuple[float, int, int, int]], wavelength: float) -> float:
    """Least-squares cubic lattice parameter from indexed peaks.

    `peaks` is a list of (two_theta_deg, h, k, l). The fit is an unweighted least squares
    through the origin, sin^2(theta) = (lambda^2 / 4a^2) * (h^2 + k^2 + l^2): no per-peak
    uncertainties, and no zero-offset term, so a displaced specimen biases the result.
    """
    xs, ys = [], []
    for two_theta_deg, h, k, l in peaks:
        n_sq = h * h + k * k + l * l
        if n_sq == 0:
            raise ValueError("(0,0,0) cannot be indexed")
        xs.append(n_sq)
        ys.append(math.sin(math.radians(two_theta_deg / 2.0)) ** 2)
    denom = sum(x * x for x in xs)
    if denom == 0:
        raise ValueError("no peaks given")
    slope = sum(x * y for x, y in zip(xs, ys)) / denom
    if slope <= 0:
        raise ValueError("degenerate peak list")
    return wavelength / (2.0 * math.sqrt(slope))
def peak_positions(hkls, wavelength: float, **cell) -> list[tuple[tuple[int, int, int], float, float]]:
```

```
    """(hkl, d, 2theta) for every reflection the wavelength can reach, sorted by 2theta."""
    out = []
    for h, k, l in hkls:
        d = d_spacing(h, k, l, **cell)
        try:
            out.append(((h, k, l), d, two_theta(d, wavelength)))
        except ValueError:
            continue
    return sorted(out, key=lambda row: row[2])
```

xrdsim/plot.py

```
"""Matplotlib rendering of a scan.

Assignment 1 already fixed the house style: a scan is drawn as counts against 2theta,
with the reflections the model predicts marked, and nothing decorated that the data do
not carry.
"""

from __future__ import annotations

import matplotlib

matplotlib.use("Agg")

import matplotlib.pyplot as plt

from .diffractometer import Scan


def plot_scan(scan: Scan, ax=None, title: str | None = None, annotate: bool = True):
    """Draw one scan; returns the axes."""
    if ax is None:
        _, ax = plt.subplots(figsize=(9, 4.2))
    ax.plot(scan.two_theta, scan.counts, lw=0.8, color="#1a3d6d")
    if annotate and scan.reflections:
        strongest = max(row.intensity for row in scan.reflections)
        placed: list[float] = []
        for row in sorted(scan.reflections, key=lambda r: -r.intensity):
            if row.intensity < 0.08 * strongest:
                continue
            if any(abs(row.two_theta - angle) < 1.2 for angle in placed):
                continue
            placed.append(row.two_theta)
            ax.annotate("".join(str(v) for v in row.hkl), (row.two_theta, 0),
                        textcoords="offset points", xytext=(0, 4),
                        fontsize=6.5, rotation=90, ha="center", va="bottom",
                        color="#8a3b12")
    ax.set_xlabel("2theta (deg)")
    ax.set_ylabel("counts")
    if title:
        ax.set_title(title)
    ax.grid(alpha=0.25, lw=0.5)
    return ax


def save_scan(scan: Scan, path, title: str | None = None, dpi: int = 160) -> None:
    """Write one scan to an image file."""
    fig, ax = plt.subplots(figsize=(9, 4.2))
    plot_scan(scan, ax=ax, title=title or (scan.phase.name if scan.phase else None))
    fig.tight_layout()
    fig.savefig(path, dpi=dpi)
    plt.close(fig)
```

xrdsim/profile.py

```
"""Peak profiles and the instrument resolution function.

The profile is normalised to unit area, so a peak's height follows from its integrated
intensity and its width. Angles are in degrees, widths are FWHM in degrees.
"""

from __future__ import annotations

import math

TWO_LN2 = 2.0 * math.log(2.0)


def gaussian(x: float, center: float, fwhm: float) -> float:
    """Unit-area Gaussian."""
    sigma = fwhm / (2.0 * math.sqrt(TWO_LN2))
    z = (x - center) / sigma
```

```
    return math.exp(-0.5 * z * z) / (sigma * math.sqrt(2.0 * math.pi))


def lorentzian(x: float, center: float, fwhm: float) -> float:
    """Unit-area Lorentzian."""
    half = fwhm / 2.0
    return (half / math.pi) / ((x - center) ** 2 + half * half)


def pseudo_voigt(x: float, center: float, fwhm: float, eta: float) -> float:
    """Unit-area pseudo-Voigt: eta of Lorentzian mixed with (1 - eta) of Gaussian."""
    if not 0.0 <= eta <= 1.0:
        raise ValueError("eta must lie in [0, 1]")
    return eta * lorentzian(x, center, fwhm) + (1.0 - eta) * gaussian(x, center, fwhm)


def caglioti_fwhm(two_theta_deg: float, U: float, V: float, W: float) -> float:
    """Instrumental FWHM in degrees: U tan^2(theta) + V tan(theta) + W."""
    t = math.tan(math.radians(two_theta_deg / 2.0))
    value = U * t * t + V * t + W
    return math.sqrt(max(value, 0.0))


def scherrer_size(fwhm_deg: float, two_theta_deg: float, wavelength: float,
                  k: float = 0.9) -> float:
    """Crystallite size in angstrom from a sample-broadened FWHM (Scherrer).

    `fwhm_deg` must already be corrected for instrumental broadening.
    """
    if fwhm_deg <= 0:
        raise ValueError("FWHM must be positive")
    beta = math.radians(fwhm_deg)
    theta = math.radians(two_theta_deg / 2.0)
    return k * wavelength / (beta * math.cos(theta))


def size_fwhm(size_a: float, two_theta_deg: float, wavelength: float,
              k: float = 0.9) -> float:
    """FWHM in degrees produced by a crystallite of `size_a` angstrom (inverse Scherrer)."""
    if size_a <= 0:
        raise ValueError("size must be positive")
    theta = math.radians(two_theta_deg / 2.0)
    return math.degrees(k * wavelength / (size_a * math.cos(theta)))


def add_in_quadrature(*fwhms: float) -> float:
    """Combine Gaussian widths."""
    return math.sqrt(sum(f * f for f in fwhms))


def pseudo_voigt_array(x, center: float, fwhm: float, eta: float):
    """Unit-area pseudo-Voigt over an array; same numbers as pseudo_voigt, vectorised."""
    import numpy as np
    if not 0.0 <= eta <= 1.0:
        raise ValueError("eta must lie in [0, 1]")
    values = np.asarray(x, dtype=float)
    sigma = fwhm / (2.0 * math.sqrt(TWO_LN2))
    half = fwhm / 2.0
    z = (values - center) / sigma
    gaussian_part = np.exp(-0.5 * z * z) / (sigma * math.sqrt(2.0 * math.pi))
    lorentzian_part = (half / math.pi) / ((values - center) ** 2 + half * half)
    return eta * lorentzian_part + (1.0 - eta) * gaussian_part
```

xrdsim/samples.py

```
"""The phase database: cell, symmetry-expanded basis and the source of each number.

Coordinates are fractions of the cell edge. Every site list below is the full basis of
the conventional cell, already expanded by the space-group operations, so the structure
factor needs no symmetry engine. Lattice constants carry the record they came from.
"""

from __future__ import annotations

from dataclasses import dataclass, field


@dataclass(frozen=True)
class Site:
    element: str
    x: float
    y: float
    z: float
```

```
    occupancy: float = 1.0
    b_iso: float = 0.5          # square angstrom; a stated assumption for every phase


@dataclass(frozen=True)
class Phase:
    key: str
    name: str
    formula: str
    space_group: str
    a: float
    c: float | None = None
    b: float | None = None
    hexagonal: bool = False
    sites: tuple[Site, ...] = field(default_factory=tuple)
    source: str = ""

    @property
    def system(self) -> str:
        if self.hexagonal:
            return "hexagonal"
        if self.b is not None:
            return "orthorhombic"
        return "cubic" if self.c is None else "tetragonal"

    def cell_kwargs(self) -> dict:
        return {"a": self.a, "b": self.b, "c": self.c, "hexagonal": self.hexagonal}

    @property
    def volume(self) -> float:
        """Conventional-cell volume in cubic angstrom."""
        if self.hexagonal:
            return self.a * self.a * self.c * 0.8660254037844386
        if self.b is None and self.c is None:
            return self.a ** 3
        return self.a * (self.b or self.a) * self.c


def _fcc(element: str, b_iso: float = 0.5) -> tuple[Site, ...]:
    return tuple(Site(element, x, y, z, b_iso=b_iso) for x, y, z in
                 [(0, 0, 0), (0, .5, .5), (.5, 0, .5), (.5, .5, 0)])


def _rocksalt(cation: str, anion: str) -> tuple[Site, ...]:
    sites = list(_fcc(cation))
    sites += [Site(anion, x + .5, y, z) for x, y, z in
              [(0, 0, 0), (0, .5, .5), (.5, 0, .5), (.5, .5, 0)]]
    return tuple(sites)


PHASES: dict[str, Phase] = {
    "si": Phase(
        key="si", name="Silicon", formula="Si", space_group="Fd-3m (227)",
        a=5.431144, sites=tuple(
            [Site("Si", x, y, z) for x, y, z in
             [(0, 0, 0), (0, .5, .5), (.5, 0, .5), (.5, .5, 0),
              (.25, .25, .25), (.25, .75, .75), (.75, .25, .75), (.75, .75, .25)]]),
        source="NIST SRM 640f certificate, a = 5.431144(8) A at 22.5 C"),
    "al": Phase(
        key="al", name="Aluminium", formula="Al", space_group="Fm-3m (225)",
        a=4.0495, sites=_fcc("Al"),
        source="room-temperature conventional value 4.0495 A; the AFLOW Al1_ICSD_52611 primitive (2.856 A) con
    verts to 4.0398 A, a different entry"),
    "cu": Phase(
        key="cu", name="Copper", formula="Cu", space_group="Fm-3m (225)",
        a=3.6149, sites=_fcc("Cu"),
        source="room-temperature conventional value 3.6149 A; AFLOW Cu1_ICSD_53755 converts to 3.6146 A"),
    "fe": Phase(
        key="fe", name="alpha-Iron", formula="Fe", space_group="Im-3m (229)",
        a=2.8665, sites=(Site("Fe", 0, 0, 0), Site("Fe", .5, .5, .5)),
        source="room-temperature conventional value; the AFLOW ICSD entry reads 2.9647 A and is flagged as exp
    anded"),
    "nacl": Phase(
        key="nacl", name="Rock salt", formula="NaCl", space_group="Fm-3m (225)",
        a=5.6402, sites=_rocksalt("Na", "Cl"),
        source="room-temperature conventional value 5.6402 A; AFLOW Cl1Na1_ICSD_60280 converts to 5.665 A"),
    "mgo": Phase(
        key="mgo", name="Periclase", formula="MgO", space_group="Fm-3m (225)",
        a=4.2112, sites=_rocksalt("Mg", "O"),
        source="room-temperature conventional value 4.2112 A; AFLOW Mg1O1_ICSD_173128 converts to 4.253 A"),
    "zno": Phase(
        key="zno", name="Zincite", formula="ZnO", space_group="P63mc (186)",
        a=3.2494, c=5.2038, hexagonal=True, sites=(
            Site("Zn", 1/3, 2/3, 0.0), Site("Zn", 2/3, 1/3, 0.5),
```

```
            Site("O", 1/3, 2/3, 0.3830), Site("O", 2/3, 1/3, 0.8830)),
        source="COD 9004180, a = 3.2494, c = 5.2038 A; u(O) = 0.3830"),
    "tio2-rutile": Phase(
        key="tio2-rutile", name="Rutile", formula="TiO2", space_group="P42/mnm (136)",
        a=4.5937, c=2.9587, sites=(
            Site("Ti", 0, 0, 0), Site("Ti", .5, .5, .5),
            Site("O", .3047, .3047, 0), Site("O", .6953, .6953, 0),
            Site("O", .8047, .1953, .5), Site("O", .1953, .8047, .5)),
        source="room-temperature conventional values a = 4.5937, c = 2.9587 A; u(O) = 0.3047. COD 9004143 is a
     600 C record, which is why its cell reads 4.616/2.977 A"),
    "tio2-anatase": Phase(
        key="tio2-anatase", name="Anatase", formula="TiO2", space_group="I41/amd (141)",
        a=3.7845, c=9.5143, sites=(
            Site("Ti", 0, 0, 0), Site("Ti", 0, .5, .25), Site("Ti", .5, .5, .5), Site("Ti", .5, 0, .75),
            Site("O", 0, 0, .2079), Site("O", 0, .5, .4579), Site("O", .5, .5, .7079),
            Site("O", .5, 0, .9579), Site("O", 0, .5, .0421), Site("O", 0, 0, .7921),
            Site("O", .5, 0, .5421), Site("O", .5, .5, .2921)),
        source="room-temperature cell a = 3.7845, c = 9.5143 A; z(O) = 0.2079 with the basis expanded by gemmi
     from COD 9008214, a 300 C record"),
}


def get_phase(key: str) -> Phase:
    try:
        return PHASES[key]
    except KeyError:
        raise KeyError(f"unknown phase {key!r}; choose from {', '.join(sorted(PHASES))}") from None
```

xrdsim/sources.py

```
"""X-ray source: the copper tube and its emission lines.

Wavelengths in angstrom. The K-alpha doublet is what a powder pattern shows; K-beta is
normally removed by a filter or a monochromator and is only modelled to show why.
"""

from __future__ import annotations

from dataclasses import dataclass

CU_KA1 = 1.540598
CU_KA2 = 1.544426
CU_KB = 1.392250

# K-alpha2 / K-alpha1 intensity ratio, and K-beta / K-alpha ratio for a Cu tube.
KA2_TO_KA1 = 0.497
KB_TO_KA = 0.20


@dataclass(frozen=True)
class Line:
    wavelength: float
    weight: float


def emission_lines(ka2: bool = True, kbeta: bool = False) -> list[Line]:
    """Emission lines of a Cu tube with their relative intensities."""
    lines = [Line(CU_KA1, 1.0)]
    if ka2:
        lines.append(Line(CU_KA2, KA2_TO_KA1))
    if kbeta:
        lines.append(Line(CU_KB, KB_TO_KA * (1.0 + (KA2_TO_KA1 if ka2 else 0.0))))
    return lines


def weighted_wavelength(lines: list[Line]) -> float:
    """Intensity-weighted mean wavelength of the lines in use."""
    total = sum(line.weight for line in lines)
    if total <= 0:
        raise ValueError("no lines")
    return sum(line.wavelength * line.weight for line in lines) / total
```

xrdsim/structure.py

```
"""Structure factors, multiplicities and systematic absences.

A phase carries its cell and an explicit, symmetry-expanded basis; this module turns
that into |F(hkl)|^2 and the multiplicity the powder average needs.
"""

from __future__ import annotations

import cmath
```

```
import math

from .atoms import debye_waller, f0
from .lattice import d_spacing


def structure_factor(phase, h: int, k: int, l: int, wavelength: float) -> complex:
    """F(hkl) in electrons, including the Debye-Waller term of every site."""
    d = d_spacing(h, k, l, **phase.cell_kwargs())
    s = 1.0 / (2.0 * d)                # sin(theta)/lambda, by Bragg's law
    total = 0j
    for site in phase.sites:
        phase_angle = 2.0 * math.pi * (h * site.x + k * site.y + l * site.z)
        total += (site.occupancy * f0(site.element, s) * debye_waller(site.b_iso, s)
                  * cmath.exp(1j * phase_angle))
    return total


def multiplicity(h: int, k: int, l: int, system: str) -> int:
    """Number of symmetry-equivalent planes in the powder average."""
    h, k, l = abs(h), abs(k), abs(l)
    if system == "cubic":
        perms = 6 if len({h, k, l}) == 3 else (3 if len({h, k, l}) == 2 else 1)
        return perms * 2 ** sum(v != 0 for v in (h, k, l))
    if system == "tetragonal":
        hk = 2 if h != k else 1
        return (hk * 2 ** sum(v != 0 for v in (h, k))) * (2 if l else 1)
    if system == "hexagonal":
        if h == 0 and k == 0:
            return 2 if l else 1
        # (h, k) on a mirror plane (h == 0, k == 0 or h == k) has a 6-member in-plane orbit;
        # a general (h, k) has 12. A non-zero l doubles either.
        in_plane = 6 if (h == 0 or k == 0 or h == k) else 12
        return 2 * in_plane if l else in_plane
    raise ValueError(f"multiplicity is not defined for {system!r}")


def lorentz_polarisation(two_theta_deg: float, monochromator_two_theta: float | None = None) -> float:
    """LP factor for an unpolarised tube, optionally with a monochromator."""
    t = math.radians(two_theta_deg)
    theta = t / 2.0
    cos2t = math.cos(t) ** 2
    if monochromator_two_theta is None:
        numerator = 1.0 + cos2t
    else:
        numerator = 1.0 + cos2t * math.cos(math.radians(monochromator_two_theta)) ** 2
    return numerator / (math.sin(theta) ** 2 * math.cos(theta))


def relative_intensity(phase, h: int, k: int, l: int, wavelength: float,
                       monochromator_two_theta: float | None = None) -> float:
    """I/I0 for one reflection: |F|^2 m LP, normalised later across the pattern."""
    f = structure_factor(phase, h, k, l, wavelength)
    d = d_spacing(h, k, l, **phase.cell_kwargs())
    two_theta = math.degrees(2.0 * math.asin(wavelength / (2.0 * d)))
    m = multiplicity(h, k, l, phase.system)
    return abs(f) ** 2 * m * lorentz_polarisation(two_theta, monochromator_two_theta)
```

Appendix B: constants and their provenance

The cell constants the report uses, with the record each came from. The full table, including confidence for every value, is data/SOURCES.md in the repository.

Key	Cell (A)	Record
si	a = 5.4311	NIST SRM 640f certificate, a = 5.431144(8) A at 22.5 C
al	a = 4.0495	room-temperature conventional value 4.0495 A; the AFLOW Al1_ICSD_52611 primitive (2.856 A) converts to 4.0398 A, a different entry
cu	a = 3.6149	room-temperature conventional value 3.6149 A; AFLOW Cu1_ICSD_53755 converts to 3.6146 A
fe	a = 2.8665	room-temperature conventional value; the AFLOW ICSD entry reads 2.9647 A and is flagged as expanded
nacl	a = 5.6402	room-temperature conventional value 5.6402 A; AFLOW Cl1Na1_ICSD_60280 converts to 5.665 A

Key	Cell (A)	Record
mgo	a = 4.2112	room-temperature conventional value 4.2112 A; AFLOW Mg1O1_ICSD_173128 converts to 4.253 A
zno	a = 3.2494, c = 5.2038	COD 9004180, a = 3.2494, c = 5.2038 A; u(O) = 0.3830
tio2-rutile	a = 4.5937, c = 2.9587	room-temperature conventional values a = 4.5937, c = 2.9587 A; u(O) = 0.3047. COD 9004143 is a 600 C record, which is why its cell reads 4.616/2.977 A
tio2-anatase	a = 3.7845, c = 9.5143	room-temperature cell a = 3.7845, c = 9.5143 A; z(O) = 0.2079 with the basis expanded by gemmi from COD 9008214, a 300 C record

Table 8: cell constants and the record behind each. Where a record is a room-temperature conventional value rather than a certified one, data/SOURCES.md marks it provisional.

Instrument constants: Cu K-alpha1 1.540598 A, K-alpha2 1.544426 A (ratio 0.497), K-beta 1.392250 A; Caglioti U = 0.004, V = -0.002, W = 0.003 deg^2; eta = 0.5; scan ceiling 150 deg; Debye-Waller B = 0.5 A^2 for every site. The form factors are the Waasmaier-Kirfel parameterisation [6], tabulated once and shipped with the code.

Appendix C: reflection tables for the nine phases

Every reflection the model reaches below 150 deg with a non-zero structure factor, with its spacing, its angle, its multiplicity and its intensity relative to the strongest line of that phase. Generated by report/tables.py; the same table is in report/tables.md.

Silicon (Si), Fd-3m (227)

hkl	d (A)	2theta (deg)	m	I/I0 (%)
111	3.1357	28.441	8	100.0
220	1.9202	47.301	12	64.3
311	1.6376	56.120	24	37.3
400	1.3578	69.127	6	9.6
331	1.2460	76.373	24	14.1
422	1.1086	88.026	24	19.4
333	1.0452	94.948	8	2.8
511	1.0452	94.948	24	8.3
440	0.9601	106.703	12	7.2
531	0.9180	114.086	48	14.0
620	0.8587	127.536	24	15.0
533	0.8282	136.883	24	8.5

Table 10: reflections of Silicon below 150 deg above 0.5 per cent of the strongest line.

Aluminium (Al), Fm-3m (225)

hkl	d (A)	2theta (deg)	m	I/I0 (%)
111	2.3380	38.473	8	100.0
200	2.0248	44.722	6	47.6
220	1.4317	65.099	12	28.5
311	1.2210	78.232	24	31.5
222	1.1690	82.439	8	9.0
400	1.0124	99.084	6	4.4

hkl	d (A)	2theta (deg)	m	I/I0 (%)
331	0.9290	112.024	24	15.6
420	0.9055	116.574	24	15.5
422	0.8266	137.462	24	18.4

Table 11: reflections of Aluminium below 150 deg above 0.5 per cent of the strongest line.

Copper (Cu), Fm-3m (225)

hkl	d (A)	2theta (deg)	m	I/I0 (%)
111	2.0871	43.318	8	100.0
200	1.8075	50.451	6	45.9
220	1.2781	74.128	12	24.2
311	1.0899	89.941	24	27.1
222	1.0435	95.151	8	8.0
400	0.9037	116.939	6	4.9
331	0.8293	136.509	24	23.5
420	0.8083	144.715	24	27.8

Table 12: reflections of Copper below 150 deg above 0.5 per cent of the strongest line.

alpha-Iron (Fe), Im-3m (229)

hkl	d (A)	2theta (deg)	m	I/I0 (%)
110	2.0269	44.672	12	100.0
200	1.4332	65.020	6	13.8
211	1.1702	82.331	24	24.8
220	1.0135	98.940	12	8.0
310	0.9065	116.376	24	14.2
222	0.8275	137.148	8	5.8

Table 13: reflections of alpha-Iron below 150 deg above 0.5 per cent of the strongest line.

Rock salt (NaCl), Fm-3m (225)

hkl	d (A)	2theta (deg)	m	I/I0 (%)
111	3.2564	27.366	8	8.2
200	2.8201	31.703	6	100.0
220	1.9941	45.447	12	63.8
311	1.7006	53.868	24	1.9
222	1.6282	56.472	8	19.9
400	1.4101	66.226	6	8.5
331	1.2940	73.069	24	0.9
420	1.2612	75.291	24	21.9
422	1.1513	83.990	24	15.7
511	1.0855	90.414	24	0.7
440	0.9971	101.171	12	5.4
531	0.9534	107.797	48	1.5

hkl	d (A)	2theta (deg)	m	I/I0 (%)
442	0.9400	110.057	24	10.1
600	0.9400	110.057	6	2.5
620	0.8918	119.484	24	10.2
533	0.8601	127.163	24	1.0
622	0.8503	129.896	24	11.2
444	0.8141	142.243	8	4.6

Table 14: reflections of Rock salt below 150 deg above 0.5 per cent of the strongest line.

Periclase (MgO), Fm-3m (225)

hkl	d (A)	2theta (deg)	m	I/I0 (%)
111	2.4313	36.942	8	10.6
200	2.1056	42.918	6	100.0
220	1.4889	62.312	12	52.5
311	1.2697	74.698	24	6.3
222	1.2157	78.639	8	14.8
400	1.0528	94.053	6	6.5
331	0.9661	105.749	24	2.8
420	0.9417	109.776	24	20.4
422	0.8596	127.302	24	20.7
333	0.8104	143.781	8	1.1
511	0.8104	143.781	24	3.2

Table 15: reflections of Periclase below 150 deg above 0.5 per cent of the strongest line.

Zincite (ZnO), P63mc (186)

hkl	d (A)	2theta (deg)	m	I/I0 (%)
100	2.8141	31.773	6	52.1
002	2.6019	34.441	2	39.9
101	2.4753	36.262	12	100.0
102	1.9104	47.558	12	22.5
110	1.6247	56.604	6	33.3
103	1.4766	62.888	12	30.6
200	1.4070	66.386	6	4.7
112	1.3781	67.968	12	26.5
201	1.3583	69.100	12	13.5
004	1.3010	72.613	2	2.3
202	1.2377	76.981	12	4.4
104	1.1809	81.433	12	2.4
203	1.0927	89.647	12	9.7
210	1.0636	92.809	12	3.3
211	1.0421	95.327	24	10.9

hkl	d (A)	2theta (deg)	m	I/I0 (%)
114	1.0155	98.670	12	6.2
212	0.9845	102.962	24	4.3
105	0.9761	104.208	12	7.9
204	0.9552	107.494	12	1.4
300	0.9380	110.409	6	5.2
213	0.9067	116.322	24	13.9
302	0.8824	121.602	12	8.2
006	0.8673	125.285	2	1.2
205	0.8367	134.029	12	8.6
106	0.8288	136.678	12	2.1
214	0.8234	138.606	24	3.4
220	0.8123	142.968	6	7.0

Table 16: reflections of Zincite below 150 deg above 0.5 per cent of the strongest line.

Rutile (TiO2), P42/mnm (136)

hkl	d (A)	2theta (deg)	m	I/I0 (%)
110	3.2482	27.436	4	100.0
101	2.4874	36.080	8	45.4
200	2.2969	39.190	4	6.6
111	2.1873	41.239	8	18.6
210	2.0544	44.043	8	6.6
211	1.6875	54.321	16	55.0
220	1.6241	56.626	4	16.4
002	1.4793	62.759	2	7.7
310	1.4527	64.047	8	7.8
221	1.4237	65.510	8	0.6
301	1.3599	69.004	8	19.2
112	1.3463	69.802	8	9.4
311	1.3040	72.418	16	1.1
202	1.2437	76.538	8	2.0
212	1.2005	79.831	16	1.1
321	1.1702	82.337	16	4.0
400	1.1484	84.249	4	2.8
410	1.1141	87.480	8	1.0
222	1.0937	89.551	8	6.7
330	1.0827	90.703	4	3.6
411	1.0427	95.255	16	6.3
312	1.0365	96.006	16	4.8
420	1.0272	97.166	8	2.8

hkl	d (A)	2theta (deg)	m	I/I0 (%)
421	0.9704	105.088	16	0.6
103	0.9643	106.041	8	2.0
113	0.9437	109.424	8	0.6
402	0.9072	116.235	8	3.5
510	0.9009	117.527	8	4.0
412	0.8900	119.887	16	1.2
213	0.8891	120.084	16	7.6
431	0.8774	122.785	16	7.1
501	0.8774	122.785	8	0.8
332	0.8737	123.678	8	5.7
422	0.8437	131.836	16	5.8
303	0.8291	136.570	8	7.4
521	0.8196	140.035	16	14.9
440	0.8121	143.091	4	2.2

Table 17: reflections of Rutile below 150 deg above 0.5 per cent of the strongest line.

Anatase (TiO2), I41/amd (141)

hkl	d (A)	2theta (deg)	m	I/I0 (%)
101	3.5165	25.307	8	100.0
103	2.4308	36.951	8	6.1
004	2.3786	37.792	2	20.1
112	2.3323	38.570	8	6.9
200	1.8922	48.043	4	28.0
105	1.7001	53.886	8	17.9
211	1.6663	55.068	16	17.4
213	1.4932	62.113	16	2.8
204	1.4808	62.689	8	13.5
116	1.3642	68.756	8	5.9
220	1.3380	70.297	4	6.6
215	1.2646	75.050	16	10.2
301	1.2506	76.044	8	2.7
303	1.1722	82.166	8	0.6
224	1.1662	82.682	8	5.0
312	1.1606	83.166	16	1.8
217	1.0598	93.248	16	0.6
305	1.0514	94.213	8	2.6
321	1.0433	95.178	16	2.9
208	1.0069	99.814	8	1.1
323	0.9965	101.253	16	0.8

hkl	d (A)	2theta (deg)	m	I/I0 (%)
316	0.9552	107.489	16	3.9
400	0.9461	109.009	4	2.2
325	0.9191	113.884	16	4.2
411	0.9136	114.941	16	2.5
228	0.8889	120.125	8	1.1
413	0.8817	121.774	16	0.8
404	0.8791	122.376	8	3.1
332	0.8767	122.948	8	0.7
420	0.8462	131.083	8	4.9
327	0.8307	136.015	16	0.7
415	0.8267	137.419	16	5.4

Table 18: reflections of Anatase below 150 deg above 0.5 per cent of the strongest line.